The Moon · Knowledge Organiser

Learning goal: I can explain how the Moon orbits the Earth and why we see different Moon phases.

Today I Can

I can state that the Moon is a rocky sphere that orbits the Earth in about 28 days.

I can explain that the Moon does not make its own light, but reflects light from the Sun.

I can explain that Moon phases happen because we see different amounts of the sunlit half as the Moon orbits Earth.

Must-Know Words

natural satellite: a body in space, like the Moon, that orbits a planet

reflect: when light from the Sun bounces off the surface of the Moon

phases of the Moon: the changing shapes of the sunlit part of the Moon as seen from Earth

waxing and waning: waxing means the lit part we see is growing; waning means it is shrinking

Core Science Steps

First, remember that the Sun always lights up the half of the Moon facing the Sun.

Second, as the Moon orbits the Earth over 28 days, our viewing angle of that sunlit half changes.

Third, we see the lit share grow (waxing) from New Moon to Full Moon, then shrink (waning) back to New Moon.

Keep the Science Clear

Distinguish between the top-down space view (half the Moon is always lit) and the Earth view (phases).

Flat diagrams draw the Sun, Earth, and Moon in one flat line, so it looks like Earth blocks the light every Full Moon; in reality the Moon's orbit is tilted by about 5 degrees.